

How Attention Shapes the Trading Behavior of Retail Investors

Interpreting Statistical Data from Behavioral Research

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Spring Semester 2026

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Introduction

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The Search Problem for Retail Investors

- ▶ Retail investors face an asymmetric universe of stocks ($U = \{1, \dots, N\}$), depending on whether they are *buying* or *selling*.
- ▶ Due to short constraints, individual investors can only sell stocks they already hold:
 $|U_{sell}| \approx 5$.
- ▶ However, during a buying decision, they must search across **thousands** of investable securities: $|U_{buy}| \approx 8,000$.
- ▶ Attention is a scarce resource and might play a crucial role in determining which stocks are purchased.

Research Question

How does attention affect the trading behavior of retail investors, particularly in the context of buying decisions where the choice set is large and complex? Does this effect differ depending on the type of signal (e.g., news, social media, Google Search) that captures investors' attention?

Attention Signals and Trading Behavior

- ▶ Barber and Odean (2008): *All That Glitters: The Effect of Attention and News on the Buying Behavior of Individual and Institutional Investors*.
- ▶ Da, Engelberg, and Gao (2011): *In Search of Attention*.
- ▶ Barber et al. (2022): *Attention-Induced Trading and Returns: Evidence from Robinhood Users*.

Gaps in the Literature

- ▶ Most studies focus on population level effects, with limited insights into individual heterogeneity in attention and trading behavior.
- ▶ Limited understanding of how different types of attention signals (e.g., news vs. social media) differentially impact trading decisions.
- ▶ No causal evidence on the mechanisms through which attention influences trading behavior (i.e., no proper experimental design).

- H1 Exposure Level:** High media exposure produces **higher** buying intention than low exposure.
- H2 Media Channel:** The type of attention signal affects buying intention; Social Media > News > Google Search.

- H3 Industry:** Buying intention differs across stock industry sectors.
- H4 Interactions:** The effect of one factor depends on the level of another — in particular, exposure amplifies the media channel effect.

Methodology, Data & Design

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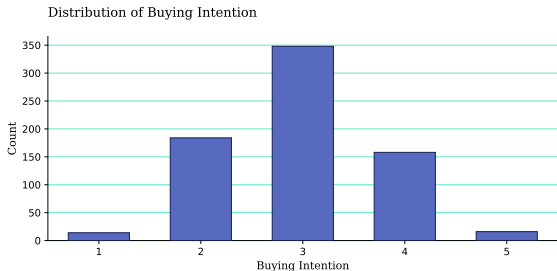
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Dataset

- ▶ $N = 720$, balanced $3 \times 2 \times 2$ factorial design
- ▶ 12 cells, 60 obs. each; no missing values
- ▶ Buying intention: $M = 2.97$, $SD = 0.80$, $Mdn = 3$

Group Sizes

Group	N
Google Search	240
News	240
Social Media	240
High Exposure	360
Low Exposure	360
Industry A	360
Industry B	360



Most responses cluster at 3; scores of 1 and 5 occur rarely, indicating adequate central spread for group comparisons.

Design

- ▶ Between-subjects $3 \times 2 \times 2$ factorial experiment
- ▶ 12 experimental conditions, 60 participants per cell
- ▶ Participants randomly assigned to exactly one condition
- ▶ Post-exposure survey: buying intention recorded immediately after the stimulus (in ideal conditions)

Variables

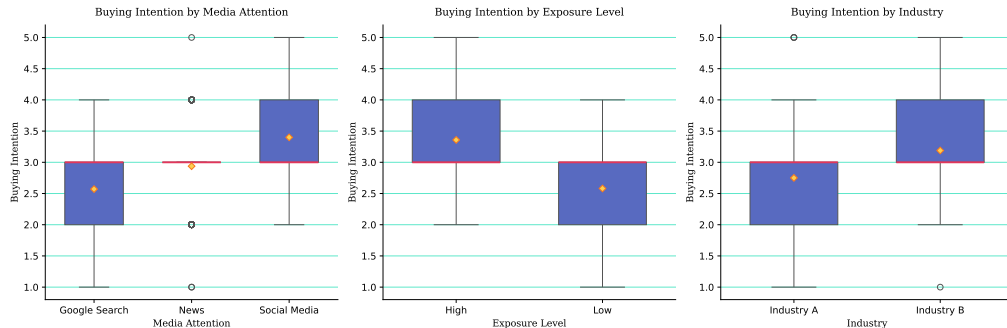
Variable	Role	Levels
Media Attention	IV1	Google Search, News, Social Media
Exposure Level	IV2	Low, High
Industry	IV3	Industry A, Industry B
Buying Intention	DV	1–5 ordinal scale

Measurement Instrument

- ▶ Structured questionnaire with channel-specific experimental stimuli:
 - ▷ Search-result interface (Google Search)
 - ▷ Formatted news article (News)
 - ▷ Social-media feed post (Social Media)
- ▶ Each stimulus presented at a low or high exposure intensity
- ▶ Buying intention measured on a 5-point Likert-type scale:
1 = very unlikely . . . 5 = very likely

Statistical Software

- ▶ **Jamovi**: preliminary tests for fast iteration
- ▶ **Python 3.13**: reproducible analysis pipeline
- ▶ `pandas` / `numpy`: data manipulation
- ▶ `statsmodels`: ANOVA, OLS regression, Tukey's HSD
- ▶ `scipy`: Shapiro-Wilk normality test
- ▶ `seaborn` / `matplotlib`: visualisations



- ▶ **Media channel:** Social Media shows the highest central tendency and widest upward spread; Google Search the lowest.
- ▶ **Exposure level:** High exposure shifts the entire distribution upward relative to Low.
- ▶ **Industry:** Participants display stronger buying intention for Industry B compared to Industry A.
- ▶ *These are preliminary visual patterns; formal statistical tests are required to confirm significance.*

Note: — red bar = median ♦ yellow diamond = mean.

Three-Way ANOVA

- ▶ Type II sums of squares — robust to unbalanced designs
- ▶ Tests all main effects and interactions simultaneously

- ▶ Partial effect size:

$$\eta_p^2 = \frac{SS_{\text{effect}}}{SS_{\text{effect}} + SS_{\text{residual}}}$$

- ▶ Post-hoc pairwise comparisons: Tukey's HSD (FWER = .05)

OLS Regression

- ▶ Model 1: main effects only
- ▶ Model 2: full model with all two-way and three-way interactions
- ▶ HC3 heteroskedasticity-robust standard errors
- ▶ Model comparison via adjusted R^2 and AIC (Akaike Information Criterion)

Diagnostics

- ▶ Shapiro-Wilk test for residual normality
- ▶ Breusch-Pagan test for heteroskedasticity
- ▶ Q-Q plot inspection

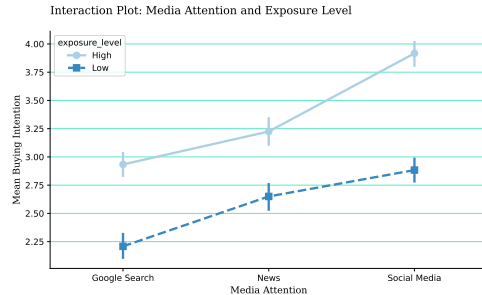
Empirical Results

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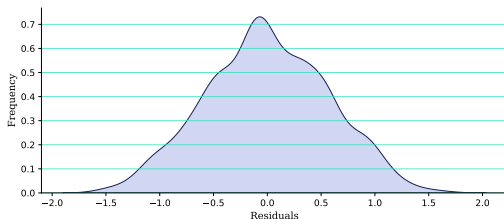
Effect	df	F	p	η_p^2
<i>Main Effects</i>				
Media Attention	2	129.04	< .001	.267
Exposure Level	1	339.12	< .001	.324
Industry	1	107.98	< .001	.132
<i>Two-Way Interactions</i>				
Media × Exposure	2	10.20	< .001	.028
Media × Industry	2	0.13	.875	.000
Exposure × Industry	1	0.28	.599	.000
<i>Three-Way Interaction</i>				
Media × Exp. × Ind.	2	1.30	.273	.004
Residual	708			

Bold = statistically significant ($p < .05$).



High exposure raises buying intention most strongly for Social Media — supporting H4.

Distribution of Residuals (ANOVA Model)

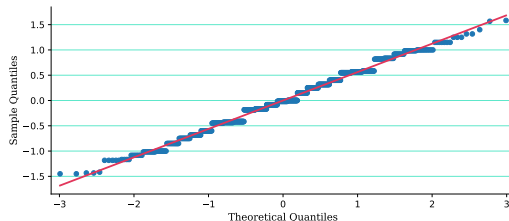


Normality

- ▶ Shapiro-Wilk: $W = 0.990$, $p < .001$
- ▶ Sensitive at large N justifies this result; in fact, minor departure only! Visual inspection is key.
- ▶ Mid-range residuals closely follow the reference line

⇒ Residuals are adequate for inference; main conclusions are robust.

Q-Q Plot of Residuals (ANOVA Model)



Homoskedasticity

- ▶ Breusch-Pagan: $LM = 4.50$, $p = .953$
- ▶ No heteroskedasticity detected
- ▶ HC3 robust SEs confirm all main effects

	Model 1 <i>Main effects</i>	Model 2 <i>Full interaction</i>
Intercept	2.74*** (0.048)	2.68*** (0.073)
News (vs. Google Search)	0.37*** (0.052)	0.33** (0.104)
Social Media (vs. Google Search)	0.83*** (0.052)	1.07*** (0.104)
Low Exposure (vs. High)	−0.78*** (0.043)	−0.68*** (0.104)
Industry B (vs. A)	0.44*** (0.043)	0.50*** (0.104)
Social Media × Low	—	−0.47** (0.146)
R^2	.491	.507
Adj. R^2	.488	.500
AIC	1246.8	1237.2

*** $p < .001$, ** $p < .01$. Standard errors in parentheses. Only significant interaction term

shown in Model 2.

Key Takeaways

- ▶ All main effects significant at $p < .001$
- ▶ Social Media adds +0.83 pts over Google Search; high exposure adds +0.78 pts over low
- ▶ Industry B adds +0.44 pts over Industry A
- ▶ Full model preferred (AIC 1237 < 1247)
- ▶ Social Media × Low: −0.47** — the channel's advantage shrinks sharply at low exposure

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Exposure Level — Strongest Effect ($\eta_p^2 = .324$)

- ▶ $F(1, 708) = 339.12, p < .001$
- ▶ High: $\bar{x} = 3.36$ vs. Low: $\bar{x} = 2.58$
($\Delta = +0.78$)
- ▶ Tukey's HSD: $p < .001$; OLS: $+0.78^{***}$

Industry ($\eta_p^2 = .132$)

- ▶ $F(1, 708) = 107.98, p < .001$
- ▶ Industry B: $\bar{x} = 3.19$ vs. A: $\bar{x} = 2.75$
($\Delta = +0.44$)
- ▶ Tukey's HSD: $p < .001$; OLS: $+0.44^{***}$

Interpretation

Consistent with the **limited attention framework**: higher visibility lowers the effective search cost and activates the *familiarity heuristic*. When a stock enters an investor's awareness, mere exposure raises perceived relevance (independently of underlying fundamentals).

Industry sector likely proxies for perceived risk and growth profile, creating heterogeneous baseline attractiveness across experimental conditions.

Media Channel Effect ($\eta_p^2 = .267$)

- ▶ $F(2, 708) = 129.04, p < .001$
- ▶ Social Media 3.40 > News 2.94 > Google Search 2.57
- ▶ All pairwise differences: $p < .001$ (Tukey's HSD)
- ▶ OLS: SM +1.07***, News +0.33** vs. Google Search

Media $\times$ Exposure Interaction ($\eta_p^2 = .028$)

- ▶ $F(2, 708) = 10.20, p < .001$
- ▶ Social Media $\times$ Low coefficient: -0.47^{**} — the channel's advantage shrinks sharply under low exposure

Interpretation

Social media activates **emotional salience**, peer-endorsement signals, and viral amplification effects absent from more passive channels (require further investigation).

The Media $\times$ Exposure interaction reveals that Social Media is inherently *exposure-sensitive*: high exposure unlocks its full potential, while low exposure narrows the gap with News. This implies that a viral social-media event carries disproportionate impact relative to a comparable uptick in search interest.

Thank You

Questions & Discussion
